

Polylogarithmic-Clock Uniformization for Phasewise Chowla-Free Lag-Two Memory

RH research program

August 8, 2026

Abstract

Fix $B > 0$. For every clock $1 \leq q \leq \lfloor (\log N)^B \rfloor$, consider the RH-379 class of universally distance-two-safe, q -periodic lag-two sign tables whose shift-two interpolation coefficient vanishes separately at each phase. We prove a limit uniform in both the clock and every table family in that class. If $S_N(q, f)$ is its normalized Möbius score and $L_q(f)$ is the fixed-clock limit, then

$$\sup_{q \leq \lfloor (\log N)^B \rfloor} \sup_{f \in \mathcal{F}_q} |S_N(q, f) - L_q(f)| \longrightarrow 0.$$

The proof truncates squarefreeness at $P = \lfloor \sqrt{\log \log N} \rfloor$. With $M_P = (\prod_{p \leq P} p)^2$, $Q = \text{lcm}(q, M_P)$, $\tau_P = \sum_{p > P} p^{-2}$, and $D_*(N)$ the maximum of the uniform Davenport sums at N and $N - 2$, we retain the explicit bound

$$|S_N(q, f) - L_q(f)| \leq \frac{4\sqrt{Q}D_*(N)}{N} + 13\tau_P + \frac{6Q}{N} + \frac{4}{N}.$$

The Fourier coefficient costs are 1, 1, 2; the last factor cannot be removed because a legal self-compatible table has $c_{21} = -2$. Consequently the finite optimizers converge uniformly to the fixed-clock optimizers, their triangular-array maximum tends to the RH-379 endpoint B_∞ , and the cofinal square clocks supply an explicit positive diagonal witness after the admissible set becomes nonempty. The result is not polynomial-clock uniformity, an adaptive-capacity limit, a projective infinite selector, an operator or trace construction, or a statement about Riemann zeros.

Keywords: Möbius function; Davenport estimate; polylogarithmic clocks; squarefree cutoff; phasewise lag-two memory.

1 Frozen problem and claim boundary

RH-366 records Davenport's estimate uniformly in the additive frequency [1, 2, 7]. RH-378 gives the exact six-monomial interpolation of lag-two tables, including the ordinary shift-two coefficient c_{11} [6]. RH-379 then optimizes the universally safe class with $c_{11}(r) = 0$ at every fixed phase and proves

$$\sup_{q < \infty} G(q) = B_\infty. \tag{1}$$

That theorem fixes q before $N \rightarrow \infty$ [4]. The purpose here is exactly one limit exchange: clocks may now grow no faster than a fixed power of $\log N$.

Let

$$\mu_0(m) = \begin{cases} \mu(m), & m \geq 1, \\ 0, & m \leq 0. \end{cases} \tag{2}$$

The zero padding in (2) is part of the finite score, not an asymptotic convention.

Definition 1.1 (Phasewise class). Fix $q \geq 1$. A family $f = (f_r)_{r \bmod q}$ belongs to $\mathcal{F}_q$ when each $f_r : \{-1, 0, 1\}^2 \rightarrow \{-1, +1\}$ and the following hold.

1. For every ternary input word, the output $\epsilon_n = f_{n \bmod q}(a_{n-2}, a_n)$ never has $\epsilon_n = \epsilon_{n+2} = +1$.
2. In the unique interpolation

$$zf_r(x, z) = c_{01}(r)z + c_{02}(r)z^2 + c_{11}(r)xz + c_{12}(r)xz^2 + c_{21}(r)x^2z + c_{22}(r)x^2z^2, \quad (3)$$

one has $c_{11}(r) = 0$ for every phase r .

The first condition is universal safety, not safety merely along the Möbius word. The second is phasewise: no cancellation between unknown shift-two correlations carrying different phase weights is assumed.

Define

$$S_N(q, f) = \frac{1}{N} \sum_{n \leq N} \mu(n) f_{n \bmod q}(\mu_0(n-2), \mu(n)). \quad (4)$$

The one-site progression densities come from RH-375, and RH-379 supplies the lag-two squarefree density and the resulting fixed-clock limit [3, 4]:

$$\delta_{q,r} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{\substack{n \leq N \\ n \equiv r \pmod{q}}} \mu(n)^2, \quad (5)$$

$$\vartheta_{q,r} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{\substack{n \leq N \\ n \equiv r \pmod{q}}} \mu(n-2)^2 \mu(n)^2. \quad (6)$$

The corresponding fixed-clock functional is

$$L_q(f) = \sum_{r \bmod q} (c_{02}(r)\delta_{q,r} + c_{22}(r)\vartheta_{q,r}). \quad (7)$$

Lemma 1.2 (Exact coefficient census). *Among the 512 truth tables, exactly 192 have $c_{11} = 0$. On those tables,*

$$\begin{aligned} c_{01}, c_{02}, c_{12} &\in \{-1, 0, 1\}, \\ c_{21}, c_{22} &\in \{-2, -1, 0, 1, 2\}. \end{aligned}$$

There are 24 distinct six-coefficient vectors, each with multiplicity eight. The table with plus-edge relation

$$E = \{(0, -1), (0, +1)\}$$

is self-compatible and has vector $(c_{01}, c_{02}, c_{11}, c_{12}, c_{21}, c_{22}) = (1, 0, 0, 0, -2, 0)$.

Proof. This is exact ternary interpolation in the basis of (3). The c_{11} histogram over all 512 tables is

c_{11}	-1	$-1/2$	0	$1/2$	1
count	32	128	192	128	32.

Exact row enumeration gives the alphabets and multiplicities. For the displayed relation, every plus edge has source 0 and target nonzero, so two plus edges cannot compose. Direct interpolation gives the stated vector. All 4608 point evaluations are reproduced independently in the finite certificate; the enumeration is a finite identity, not asymptotic evidence. $\square$

2 Squarefree cutoff and periodic Fourier bounds

Fix an integer $P \geq 2$ and put

$$\eta_P(m) = \prod_{p \leq P} (1 - \mathbf{1}_{p^2|m}), \quad M_P = \left(\prod_{p \leq P} p \right)^2, \quad \tau_P = \sum_{p > P} \frac{1}{p^2}. \quad (8)$$

Divisibility in (8) is defined for every integer. In particular,

$$\eta_P(0) = 0, \quad \eta_P(-1) = 1. \quad (9)$$

The function η_P has period M_P . This is a valid period; it need not be the minimal period of a particular product involving η_P .

Lemma 2.1 (Tail union bound). *For every positive integer m ,*

$$0 \leq \eta_P(m) - \mu(m)^2 \leq \sum_{p > P} \mathbf{1}_{p^2|m}. \quad (10)$$

Consequently, for every $X \geq 1$,

$$\sum_{m \leq X} (\eta_P(m) - \mu(m)^2) \leq \sum_{p > P} \left\lfloor \frac{X}{p^2} \right\rfloor \leq X\tau_P. \quad (11)$$

The same inequalities compare every limiting periodic squarefree mean with its P -cutoff mean.

Proof. If $\eta_P(m) = 1$ but $\mu(m)^2 = 0$, a prime $p > P$ has $p^2 \mid m$. This proves (10); summation gives (11). Notice that the exact count is $\lfloor X/p^2 \rfloor$, so no $\pi(\sqrt{X})$ or $X^{-1/2}$ loss appears. Divide the identical inequalities by X and take limits along any fixed periodic phase selection to obtain the last assertion. $\square$

For a Q -periodic complex function w , use the normalized transform

$$\hat{w}(a) = \frac{1}{Q} \sum_{r \bmod Q} w(r) e^{-2\pi i ar/Q}. \quad (12)$$

Lemma 2.2 (Periodic Möbius channel). *Let*

$$D(X) = \sup_{\alpha \in \mathbb{R}} \left| \sum_{n \leq X} \mu(n) e^{2\pi i \alpha n} \right|.$$

If w has period Q , then

$$\left| \sum_{n \leq X} w(n) \mu(n) \right| \leq \sqrt{Q} \|w\|_{\infty} D(X). \quad (13)$$

Proof. Fourier inversion and the definition of $D(X)$ give the right side with $\sum_a |\hat{w}(a)|$ in place of $\sqrt{Q} \|w\|_{\infty}$. Parseval and Cauchy–Schwarz give

$$\sum_a |\hat{w}(a)| \leq \sqrt{Q} \left(\sum_a |\hat{w}(a)|^2 \right)^{1/2} = \sqrt{Q} \left(\frac{1}{Q} \sum_r |w(r)|^2 \right)^{1/2} \leq \sqrt{Q} \|w\|_{\infty}.$$

The sup norm is essential. For $P = 2$, $q = 1$, and the legal $c_{21} = -2$ witness, one period of the relevant shifted mask is $(-2, -2, 0, -2)$. Its normalized Fourier ℓ^1 norm is 3, which is larger than $\sqrt{4} = 2$ but no larger than $2\sqrt{4}$. $\square$

3 The uniform analytic ledger

For $N \geq 3$, set

$$D_*(N) = \max_{X \in \{N, N-2\}} D(X), \quad Q = \text{lcm}(q, M_P). \quad (14)$$

Every coefficient sequence $c_{ij}(n \bmod q)$ and every cutoff mask in the proof is Q -periodic. Again, Q is only asserted to be a common period.

Proposition 3.1 (Exact conservative ledger). *For every $N \geq 3$, $P \geq 2$, $q \geq 1$, and $f \in \mathcal{F}_q$,*

$$\boxed{|S_N(q, f) - L_q(f)| \leq \frac{4\sqrt{Q}D_*(N)}{N} + 13\tau_P + \frac{6Q}{N} + \frac{4}{N}}. \quad (15)$$

Proof. Insert (3) into (4). The c_{11} channel is absent phase by phase. We account for the remaining five channels separately.

The c_{01} channel is a Q -periodic weight of sup norm one against $\mu(n)$. Lemma 2.2 bounds it by $\sqrt{Q}D(N)/N$.

For c_{12} , the first two sites vanish and the change of variables $m = n - 2$ gives

$$\sum_{m \leq N-2} c_{12}(m+2)\mu(m)\mu(m+2)^2.$$

Replace $\mu(m+2)^2$ by $\eta_P(m+2)$. The cost is at most $N\tau_P$ by Lemma 2.1. The remaining periodic weight has sup norm one, so its Fourier cost is $\sqrt{Q}D(N-2)$.

For c_{21} , replace $\mu(n-2)^2$ by $\eta_P(n-2)$ for $n \geq 3$. The coefficient bound $|c_{21}| \leq 2$ makes the tail cost $2N\tau_P$. Extend the periodic Möbius sum to $1 \leq n \leq N$. At $n = 2$, $\eta_P(0) = 0$; at $n = 1$, the cutoff has $\eta_P(-1) = 1$ whereas $\mu_0(-1) = 0$, costing at most two. The periodic weight has sup norm two. This channel therefore contributes

$$2\sqrt{Q}D(N) + 2N\tau_P + 2.$$

The three Fourier costs are thus $1 + 1 + 2 = 4$.

It remains to compare the two square-only channels with (7). For a bounded Q -periodic sequence a , complete periods plus the final incomplete block give

$$\left| \frac{1}{N} \sum_{n \leq N} a(n) - \frac{1}{Q} \sum_{r \bmod Q} a(r) \right| \leq \frac{2Q\|a\|_\infty}{N}. \quad (16)$$

For $c_{02}\mu(n)^2$, finite-to-cutoff and limit-to-cutoff each cost τ_P , while (16) costs $2Q/N$. Its total is

$$2\tau_P + 2Q/N. \quad (17)$$

For $c_{22}\mu_0(n-2)^2\mu(n)^2$, the product difference is bounded by the sum of the two one-site differences. Since $|c_{22}| \leq 2$, finite-to-cutoff costs $4\tau_P$ and limit-to-cutoff costs another $4\tau_P$. The cutoff product has sup norm two, so (16) costs $4Q/N$. Extending the cutoff product across the padded sites costs two at $n = 1$ and zero at $n = 2$ by (9). This channel contributes

$$8\tau_P + 4Q/N + 2/N. \quad (18)$$

Collecting the tail coefficients gives

$$\underbrace{1+2}_{\text{Fourier masks}} + \underbrace{2}_{c_{02}} + \underbrace{8}_{c_{22}} = 13.$$

The periodic discrepancies give $2 + 4 = 6$, and the two endpoint costs give $2 + 2 = 4$. After division by N , this is (15). $\square$

The decomposition also explains the tail constant in a second way. The finite mask replacements cost $1 + 1 + 2 + 4 = 8$ across the $c_{12}, c_{02}, c_{21}, c_{22}$ channels, and the two limiting square means cost $1 + 4 = 5$. These are conservative uniform costs; no sign cancellation between channels is used.

4 Polylogarithmic-clock uniformization

All logarithms in the clock budget are natural. For fixed $B > 0$, put

$$H_B(N) = \lfloor (\log N)^B \rfloor. \quad (19)$$

All suprema and maxima using this budget are understood for sufficiently large N , when $H_B(N) \geq 1$.

Theorem 4.1 (Uniform polylogarithmic-clock limit). *For every fixed real $B > 0$,*

$$\sup_{1 \leq q \leq H_B(N)} \sup_{f \in \mathcal{F}_q} |S_N(q, f) - L_q(f)| \longrightarrow 0. \quad (20)$$

Proof. Take

$$P = P_N = \lfloor \sqrt{\log \log N} \rfloor, \quad (21)$$

which is at least two for all sufficiently large N . The elementary estimate

$$\log M_P = 2 \sum_{p \leq P} \log p \leq 2P \log P = o(\log \log N) \quad (22)$$

shows $M_P = (\log N)^{o(1)}$. Uniformly for $q \leq H_B(N)$,

$$Q \leq qM_P \leq (\log N)^{B+o(1)}. \quad (23)$$

RH-366 freezes Davenport's estimate: for every fixed $A > 0$,

$$\sup_{\alpha \in \mathbb{R}} \left| \sum_{n \leq X} \mu(n) e^{2\pi i \alpha n} \right| \ll_A X (\log X)^{-A}. \quad (24)$$

Choose one fixed $A > B/2$. Equations (23) and (24) make the Fourier term in (15)

$$\ll_A (\log N)^{B/2-A+o(1)} = o(1).$$

Also

$$\tau_P \leq \sum_{n > P} n^{-2} \leq \frac{1}{P-1} \longrightarrow 0, \quad Q/N \longrightarrow 0, \quad 1/N \longrightarrow 0.$$

The right side of (15) is independent of f and otherwise depends on q only through the displayed uniform bound for Q . This proves (20). $\square$

The fixed nature of B is load-bearing. The proof does not supply a constant uniform in a varying $B = B(N)$. For $q = N^\varepsilon$, the $\sqrt{Q}$ Fourier mass is polynomial, while (24) supplies only an arbitrarily fixed logarithmic saving; this route therefore stops before polynomial clocks.

5 Triangular optimizers and a diagonal witness

Because every $\mathcal{F}_q$ is finite, define

$$G_N(q) = \max_{f \in \mathcal{F}_q} |S_N(q, f)|, \quad G(q) = \max_{f \in \mathcal{F}_q} |L_q(f)|. \quad (25)$$

The second quantity is exactly the RH-379 fixed-clock optimizer.

Corollary 5.1 (Uniform optimizer transfer). *For every fixed $B > 0$,*

$$\sup_{1 \leq q \leq H_B(N)} |G_N(q) - G(q)| \longrightarrow 0. \quad (26)$$

Proof. For any two finite families of real numbers (a_f) and (b_f) ,

$$|\max_f |a_f| - \max_f |b_f|| \leq \max_f |a_f - b_f|.$$

Apply this at each q and then Theorem 4.1. □

Theorem 5.2 (Restricted triangular-array endpoint). *For every fixed $B > 0$,*

$$\boxed{\max_{1 \leq q \leq H_B(N)} G_N(q) \longrightarrow B_\infty.} \quad (27)$$

Proof. Equation (1) gives $G(q) \leq B_\infty$ for every fixed q . Conversely, for every $\varepsilon > 0$ there is one fixed clock q_ε with $G(q_\varepsilon) > B_\infty - \varepsilon$. Since $H_B(N) \rightarrow \infty$, that clock is eventually admissible. Hence

$$\max_{q \leq H_B(N)} G(q) \longrightarrow B_\infty.$$

Corollary 5.1 transfers this scalar maximum to G_N . □

There is also an explicit cofinal witness, with a necessary empty-set convention. Let $3 = p_1 < p_2 < \dots$ be the odd primes and set

$$q_y = 4 \prod_{i \leq y} p_i^2. \quad (28)$$

When $H_B(N) < q_1 = 36$, record `no_square_clock_available`; do not silently substitute q_1 . Once the set is nonempty, define

$$y_B(N) = \max\{y \geq 1 : q_y \leq H_B(N)\}. \quad (29)$$

For each y , choose the RH-379 positive optimizer $f_y^+ \in \mathcal{F}_{q_y}$ with $L_{q_y}(f_y^+) = G(q_y) > 0$.

Corollary 5.3 (Square-clock diagonal). *After (29) becomes defined,*

$$S_N(q_{y_B(N)}, f_{y_B(N)}^+) \longrightarrow B_\infty. \quad (30)$$

Proof. For every fixed y , eventually $q_y \leq H_B(N)$; hence $y_B(N) \rightarrow \infty$. RH-379 proves $G(q_y) \rightarrow B_\infty$. Therefore

$$\begin{aligned} |S_N(q_{y_B}, f_{y_B}^+) - B_\infty| &\leq \sup_{q \leq H_B(N), f \in \mathcal{F}_q} |S_N(q, f) - L_q(f)| \\ &\quad + |G(q_{y_B}) - B_\infty| \longrightarrow 0. \end{aligned}$$

□

The families $f_{y_B(N)}^+$ form a triangular-array witness only. No projective compatibility across clocks, and no single infinite selector, is asserted.

6 Finite certificate, source lock, and reproducibility

The standard-library artifact reproduces the finite interfaces used above. It contains all 512 truth rows and all 4608 interpolation evaluations; the 192 zero- c_{11} rows reduce to 24 coefficient vectors of multiplicity eight. It checks maximum coefficient-vector ℓ^1 norm 3, maximum squared ℓ^2 norm 5, the self-compatible $c_{21} = -2$ witness, and the three normalized-DFT channel costs 1, 1, 2.

The cutoff rows have exact periods

$$4, \quad 36, \quad 900, \quad 44100$$

for $P = 2, 3, 5, 7$. Coprime and noncoprime least-common-multiple fixtures are included. One additional fixture has $q = 5$, $P = 3$, and $Q = 180$, while its relevant mask has minimal period 36; this prevents the valid common period from being promoted to a minimal-period claim. Exact square-mask means, the $13 = 8 + 5$ tail split, the 4 endpoint cost, and small-clock finite max-plus rows are regenerated from source.

The certificate is explicitly labelled `reproduction_not_analytic_proof`. Its finite rows do not prove Davenport’s theorem, a uniform asymptotic, or any limit in this paper. A closed official Draft 2020–12 schema fixes every nested member and exact JSON primitive type. Twenty-four nontrivial certificate mutations test coefficient, period, shift, factor-two, square-mean, tail, padding, clock, sentinel, and claim-boundary failures. Separate tests mutate source-lock commits and group digests. Optimized Python mode and bool-for-int substitutions are also tested.

The source lock contains 67 unique files in groups 51/8/8: the released RH-384 immutable closure, the standard eight RH-384 theorem files, and the standard eight RH-366 Davenport files. Every live file is compared with its Git release blob. Mutable root policy and handoff files are excluded. The RH-366 release is 0396fab97bbe3348c8237f8734dec0e1893fd3bf. Its frozen `main.tex` SHA-256 is

7df165bd63d43f52dc217dea6691d231d8e40c00c148ab7e1aa4abcac55060fb

The RH-379 release is 9ae9802ed17529ef4adfb81d7e2158d47c3c8d22. Its frozen `main.tex` SHA-256 is

c5d97a227398a4f1d46a39fdec73ffb86aeb9bfc0f16296be7023b187b497090

7 Context, limitations, and Gates

The cubic residual limits recorded in RH-384 are useful context, but they are not the discovery claim here [5]. The new theorem is the restricted growing-clock uniformization in Theorem 4.1.

Route A is `G0`. Proposition 3.1, the fixed- B cutoff, optimizer transfer, endpoint theorem, and diagonal witness form a closed theorem chain. Route B is `STOP_SCOPED`. The proof gives no polynomial-clock uniformity: a fixed logarithmic Davenport saving does not pay polynomial Fourier mass. If $c_{11}(r)$ is activated, already $q = 1$ exposes the ordinary shift-two Chowla sum, which the frozen sources do not cancel.

The following boundaries are mandatory.

1. $B > 0$ is fixed before $N \rightarrow \infty$; no $B = B(N)$ theorem is proved.
2. No effective finite threshold or rate is claimed. Constants in Davenport’s estimate depend on the chosen fixed exponent.
3. The maximum in (27) is restricted to $q \leq H_B(N)$; it is not $\sup_{q < \infty} G_N(q)$.
4. The theorem optimizes periodic phase tables, not arbitrary adaptive sign words. It supplies no upper bound proving a limit for K_N/N .
5. The diagonal witness is not a projectively compatible infinite selector.

Gates A–E remain false/open. These externally prescribed arithmetic tables do not construct a canonical intrinsic dynamical spectral determinant, time-oriented scattering completion, self-adjoint generator with an intrinsic $T \log T$ law, signed von-Mangoldt weighted prime-power trace, or completed-zeta divisor equality. Nothing here identifies Riemann zeros, constructs a Hilbert–Pólya operator, or proves the Riemann hypothesis.

8 Conclusion

The fixed-clock RH-379 limit is uniform throughout every fixed polylogarithmic clock window. The proof pays every source of growth: four units of Fourier mass, thirteen units of squarefree-tail loss, six units of period discrepancy, and four zero-padding endpoints. The cutoff period is small enough to fit inside Davenport’s logarithmic saving precisely because B is fixed. This establishes a genuine, restricted growing-clock theorem while leaving polynomial clocks, active shift-two correlations, adaptive capacity, and all spectral Gates open.

Reproducibility and declarations

Data availability. All source code, exact outputs, tests, closed schema, source hashes, and archive-verification files are included in the RH-385 repository directory. No new external dataset was created or analyzed. Finite certificate rows reproduce interfaces and are not the analytic proof.

Ethics. This mathematical study uses no human participants, animals, personal data, or biological materials. Ethics approval was not required.

Author contributions. The RH research program performed conceptualization, methodology, formal analysis, software, validation, and writing—original draft and review/editing.

Funding. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Declaration of interests. The authors declare no known competing financial interests or personal relationships that could have influenced the work reported here.

AI-assisted workflow. The manuscript and executable checks were prepared with AI-assisted academic writing and coding tools under the repository’s documented multi-agent workflow. The mathematical claims are bounded by frozen sources, explicit proofs, independent audits, and reproducible artifacts. The authors retain responsibility for the final content and its integrity.

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